QINGYU ZHANG (Alfred)

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Summary

I am a PhD researcher at University College London (UCL), funded by EPSRC and the China Scholarship Council (CSC), specialising in robotics, embodied intelligence, and bioelectronic systems. My research spans multimodal perception, tactile sensing, reinforcement learning, and human–robot interaction, with an emphasis on enabling adaptive and generalisable robotic manipulation. My current research explores the evaluation and generalisation of reusable robotic skills, alongside the joint design of high-level decision-making and low-level learning-based control for complex manipulation tasks.
I have a strong background in hardware–software co-design, embedded systems, and robotic system integration, with research experience in medical robotics and haptic interfaces. Prior to joining UCL, I graduated top of my MSc class at the University of Bristol and co-invented six patents in intelligent hardware and sensing technologies.

EDUCATION

PhD in Electronic and Electrical Engineering, UCL **Mar 2024 – Now**

MSc in Advanced Microelectronic System Engineering (AMSE), UoB **Sep 2022 - Sep 2023**

- Degree Class: **Distinction** (1st place in the class of years 22-23, overall score 83/100)
- Final project mark: 88/100

***BEng in Optical Electronic Information Engineering*, Huazhong University of Science and Technology** **Sep 2016 - Jul 2020**

- Overall grade: GPA 3.22/4
- Final project mark: 88/100

WORK EXPERIENCE

***Senior IoT Engineer*, New Oriental Education Technology Group (XDF), Beijing, China** **Feb 2021 - Dec 2021**

- Engineered a scalable and configurable full-stack controller on the ESP32 platform for smart classroom systems. Implemented BLE, audio, IMU, and multi-button peripheral integration; contributed to manufacturing of 200+ units.
- Led the development of an AI-enabled edge device for smart classroom instruction, integrating embedded sensing, on-device model inference, and real-time feedback. Supervised three IoT engineers in system design, ML workflow implementation, and modular firmware development.

***IoT Engineer*, Shensi Shudun Technology, Co, Ltd, Beijing, China** **Jul 2020 - Jan 2021**

- Developed Windows-based desktop applications using PyQt6 to streamline and automate assembly-line operations.
- Led the design and implementation of a smart dock for existing air purifiers on STM32 and ESP8266 platforms, featuring one-click network configuration and multi-protocol communication (Wi-Fi, UART, I²C); the system is mass-produced with 1,000+ units annually.
- Optimized the hardware communication architecture on the M263KI chip by extending SPI communication from four to six serial channels, significantly improving data throughput and system efficiency.

SKILLS

- **Programming & Machine Learning:** Python, C/C++, PyTorch, MATLAB, Verilog; computer vision, deep learning, and reinforcement learning.
- **Robot Learning & Embodied AI:** Robotic policy learning and evaluation, skill-based manipulation, hierarchical decision-making, multimodal perception, and vision-language-action models (research interests).
- **Robotics & Simulation:** MuJoCo, MuJoCo Warp, UR collaborative robots, Haply Inverse3 haptic interfaces, and custom robotic platforms.
- **Multimodal Sensing & Bioelectronics:** Tactile and force sensing, bioimpedance measurement, sensor fusion, and real-time acquisition systems.
- **Embedded Systems & Hardware:** ESP32, STM32, embedded C/C++, RTOS, BLE, SPI, I²C, UART, PCB design, mixed-signal circuit integration, and hardware debugging.
- **Modelling & Engineering:** COMSOL Multiphysics, SolidWorks, Blender, finite element analysis, and electrical field simulation.
- **Research and Research Writing Skills**

PUBLICATIONS

Paper:

- **Q. Zhang**, A. Al-Hindawi, A. Demosthenous, and Y. Wu, "Bioimpedance-Informed Haptic Needle System with Human-in-the-Loop Control for Central Venous Access," *IEEE/ASME Transactions on Mechatronics (TMECH)*, under review.
- J. Li, D. Jiang, N. Neshatvar, **Q. Zhang**, Y. Wu, and A. Demosthenous, "A Neural Stimulator ASIC With an Improved Blocking Capacitor Charge Balancing Method and Speed-Boosting Feedback," *IEEE Transactions on Circuits and Systems I: Regular Papers (TCAS-I)*, 2026.

Conference Paper:

- J. Li, **Q. Zhang**, D. Jiang, S. Ha, A. Demosthenous, and Y. Wu, "A 0.62 µW/sensor 82 fps Time-to-Digital Impedance Measurement IC with Unified Excitation/Readout Front-End for Large-Scale Piezo-Resistive Sensor Array," *IEEE International Solid-State Circuits Conference (ISSCC)*, 2026.
- **Q. Zhang**, A. Al-Hindawi, A. Demosthenous, and Y. Wu, "Haptic-Enhanced Bioimpedance Needle for Precision Navigation in Central Venous Catheterisation with Millimetre Accuracy," *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2025.
- **Q. Zhang**, A. Al-Hindawi, A. Demosthenous, and Y. Wu, "Live Demonstration: Haptic-Enhanced Bioimpedance Needle for Assisting Central Venous Catheterisation," *IEEE ISCAS*, 2025. **Best Demo Award.**

Additional Publications: 5 additional conference papers and live demonstrations at *IEEE ISCAS* (2025–2026), covering multimodal smart insoles, biomechanical sensing, human–machine interaction, and capsule localisation.

PATENTS

- 'Image processing method and apparatus, electronic device and computer-readable storage medium', CN115393922A, 2022
- 'Data processing methods, apparatus, storage media and electronic devices', CN114969790A, 2022
- 'Attitude recognition method, apparatus, storage medium and electronic device', CN114782994A, 2022
- 'Sitting posture recognition method, device and storage medium', CN114821652A, 2022
- 'Face recognition method, apparatus, storage medium and electronic device', CN114998955A, 2022
- 'Image processing method, apparatus, storage medium and electronic device', CN114913181A, 2022

AWARDS

- Best live demo, ISCAS, 2025
- Outstanding Employee of the Quarter in 2021, top 1.5% of 200 colleagues in the AI research institute of XDF
- Outstanding graduate from HUST in BEng and BA college in 2020
- Academic Progress Scholarship in HUST in 2019
- Outstanding Student Cadres Scholarship in HUST in 2018 and 2019
- Scientific and Technological Innovation Scholarship in HUST in 2018
- Third prize in the Mathematical Modelling Competition in central China in 2018, the top 5% of 2000 groups
- Second prize at HUST Entrepreneur Camp in 2018, second out of 26 teams

RESEARCH & PROJECTS

PhD Research: Skill-Aware Evaluation and Reward Learning for Robot Foundation Models | 2026 – Present

- Developing intent-aware skill evaluation and reward models to assess task progress, skill competence, and failure causes, guiding RL post-training and improving VLA generalisation.

PhD Research: Adaptive World Models for Long-Horizon Robotic Manipulation | 2026 – Present

- Investigating adaptive visuo-tactile world models that continuously update physical predictions from real-world interaction feedback, enabling counterfactual planning and robust decision-making in long-horizon manipulation.

Research assistant: Developing the smart insole related hardware and software system **Mar 2025 – Sep 2025**

Research assistant: Developing the ultra-low power embedded system for in vivo stimulation **Oct 2024 – Apr 2025**

PhD research: Bioimpedance needle trajectory system **May 2024 - Oct 2024**

- Developed a human-in-the-loop robotic needle guidance system integrating ultrasound segmentation, RGB-based spatial registration, bioimpedance sensing, and haptic shared control.
- Evaluated the system with five participants across four assistance conditions in phantom-based needle insertion experiments.
- **Key Results:** Achieved a 96.7% vessel-entry success rate (29/30 trials), compared with 50% for freehand insertion, reducing mean targeting error from 9.91 mm to 2.48 mm (RMSE: 2.92 mm).
- Demonstrated improved insertion-depth consistency through bioimpedance-informed haptic feedback, reducing depth-error standard deviation from 3.22 mm to 1.87 mm ($p = 0.001$).

MSc project: design and modelling of power conditioning circuits for triboelectric nanogenerator **May 2023 - Sep 2023**

- Performed finite element analysis in COMSOL to model and analyze a triboelectric nanogenerator (TENG).
- Designed an analytical TENG model in Verilog-AMS using Cadence.
- Developed a reliable and efficient power management circuit for a bladder actuator based on the TENG models.
- **Key result:** The system generates 3.25 mAh of energy per day with a 23% conversion efficiency, outperforming most designs.